Smart Traffic Coordination: AI-Based Multi-Intersection Traffic Management

Official Project & Technical Handover Document

Project Title: AI-Based Multi-Intersection Traffic Management along the Silk Board Corridor
Corridor Focus: Bengaluru Hosur Road Arterial ($J_1 \rightarrow J_2 \rightarrow J_3$ J1 → J2 → J3)
Target Audience: Government Officials, Academic Evaluators, and Traffic Police Authorities
Author / Module Lead (AI & Decision Layer): Senior AI/ML Engineer (Pallavi)
Collaborators: Sneha (SUMO Simulation & TraCI), Suma (FastAPI Backend, SQLite DB & UI)

1. Executive Summary

Urban traffic congestion at Bengaluru's **Central Silk Board** and its feeder junctions along **Hosur Road** is one of the most severe bottlenecks in India, causing severe delays, elevated vehicle emissions, and delayed emergency response times.

This project delivers an integrated, real-time **Adaptive Traffic Signal Coordination and Multi-Agent AI Decision Layer** evaluated using microscopic simulation (Eclipse SUMO) and visualized via a command-center dashboard.

Key Empirically Verified Benchmark Results (5-Seed Simulation)

Compared against the fixed-time baseline currently standard across Indian urban junctions:

- **Corridor Travel Time:** Reduced from **91.05s** to **57.50s** (**36.8% faster progression**).
- **Corridor Total Waiting Time:** Reduced from **779.43s** to **296.88s** (**61.9% reduction in delays**).
- **Corridor Average Speed:** Increased from **27.85 km/h** to **36.40 km/h** (**+30.7% progression speed**).
- **Vehicle CO_2 CO2 Emissions:** Reduced from **53.72 kg** to **42.67 kg** (**20.6% cut**).
- **Fuel Consumption:** Reduced from **17,391.65 L** to **13,812.75 L** (**20.5% fuel saved**).
- **Emergency Ambulance Traversal:** Reduced from **41.0s** to **33.6s** (**18.0% faster emergency clearance**).

2. System Architecture & Information Pipeline

The solution connects three decoupled layers to guarantee real-time reactivity, algorithmic safety, and strict technical integrity:

3. Detailed Walkthrough of Dashboard Features

The dashboard is structured into a command-center interface specifically designed for transport officials and academic evaluators:

3.1. Telemetry Health & Header Ribbon

- **Telemetry Freshness Badge (`Feed: Live (1s ago)`):** Dynamically computes elapsed seconds since the last telemetry packet. Distinguishes live streams (< 5< 5s, green), recent streams ($5 - 15$5 – 15s, amber), and stale or offline connections (> 15> 15s, red) to prevent presenting stale data as active readings.
- **Stream Status (`Stream: CONNECTED`):** Monitors the persistent WebSocket connection (`/ws/traffic`) communicating simulation telemetry directly to the frontend.
- **SUMO Core Status (`SUMO: Mock / Simulation Active`):** Reports whether the underlying simulation is operating via live TraCI or mock deterministic telemetry.
- **Simulation State (`RUNNING` / `PAUSED` / `STOPPED`):** Displays real-time operational status of the traffic coordination loop.
- **Operator Manual Override Button:** Opens an authenticated dialog enabling an operator to intervene on any junction (J_1 J1, J_2 J2, or J_3 J3) and force an emergency green or yellow clearance.

3.2. Overview KPIs Ribbon (7 Core Metrics)

All metrics strictly differentiate between **measured zero** and **missing/unreceived telemetry** (using em dashes `–` until verified readings arrive):

1. **Intersections (**$3/3$**3/3):** Scope of the Hosur Road Silk Board continuous corridor.
2. **Active Vehicles (**38**38):** Real-time count of vehicles currently progressing between J_1 J1 and J_3 J3 .
3. **Throughput (**101**101):** Number of vehicles that completed their journey across all three junctions.

4. **Average Speed (**$36.4\ \mathrm{km/h}$**36.4 km/h):** Progression speed along Hosur Road ($+30.7\%$+30.7% over baseline).
5. **Total Waiting Time (**$296.9\mathrm{s}$**296.9s):** Cumulative corridor idle queue delay (-61.9%−61.9% reduction vs. baseline), with the calculated per-vehicle average displayed as a subtitle ($7.8\ \mathrm{s/veh}$ 7.8 s/veh).
6. $\mathrm{CO_2}$ **CO2 Emissions (**$42.7\ \mathrm{kg}$**42.7 kg):** Cumulative greenhouse gas emissions, reflecting a 20.6% 20.6% reduction. Subtitle displays cumulative $\mathrm{NO_x}$ NOx ($184.2\ \mathrm{g}$184.2 g).
7. **Fuel Consumed (**$13,812\ \mathrm{L}$**13,812 L):** Calculated via the microscopic HBEFA 3.1 vehicle emission model, showing 20.5%20.5% fuel conservation.

3.3. Simulation Control Deck

- **Simulation Time Progress:** Displays elapsed simulation seconds and total requested horizon (e.g., $145.0\mathrm{s/600s}$145.0s/600s, 24%24%).
- **Controller Mode Selector:** Allows selecting between:
  - *Coordinated Adaptive Controller (Silk Board Inflow Anticipation)*
  - *Fixed-Time Baseline (Static Uncoordinated Timings)*
  - *MARL PPO Prototype*
  - *MARL DQN Prototype*
- **Parameter Controls:** Duration dropdown ($300\mathrm{s}$300s, $600\mathrm{s}$600s, $1200\mathrm{s}$1200s), random traffic seed selector ($42$42, $101$101, etc.), and emergency preemption checkbox.
- **Interactive Lifecycle Buttons:** `Pause`, `Resume`, `Stop`, and `Reset` directly control the background simulation engine without page reloads.

3.4. Silk Board Directional Corridor Topology

Replaces vertical stacking with a horizontal, directional progression diagram:

- **Left Node (**J_1 **J1 - Hosur Rd North Feeder):**
  - Traffic Light ID: `1617743335`
  - Inflow Edge: `172853382#2` (Hosur North)
  - Side Cross Street: `-148697299_0`
  - Signal State: Active Green (P_2 P2) with vehicle count ($12$12), queue ($2\ \mathrm{veh}$2 veh), and average speed ($38.2\ \mathrm{km/h}$38.2 km/h).
- **Connecting Arterial Link (**$J_1 \rightarrow J_2$ **J1 → J2):**
  - SUMO Edge: `172853382#3` ($400\mathrm{m}$400m Link)

    - Directional flow arrows (`➔ ➔ ➔`) with real-time platoon indicators.
- **Center Node (J_2 J2 - Central Silk Board Primary Hub):**
    - Traffic Light ID: `GS_cluster_10282769895...`
    - Approaches: Hosur Corridor Inflow (`172853382#3`) and Madiwala Cross Feeder.
    - Signal State: Active Green (P_0 P0) with vehicle count (1818), corridor queue ($5\ \text{veh}$5 veh), Madiwala queue ($3\ \text{veh}$3 veh), and speed ($34.6\ \text{km/h}$34.6 km/h).
    - **Inflow Anticipation Active:** Anticipates vehicle arrival times from J_1 J1 to proactively hold green and clear platoons before congestion accumulates.
- **Connecting Southbound Link ($J_2 \rightarrow J_3$ J2 → J3):**
    - SUMO Edge: `1148717038#0` (350m350m Link)
    - Directional flow arrows (`➔ ➔ ➔`).
- **Right Node (J_3 J3 - Hosur Rd South Feeder):**
    - Traffic Light ID: `6970466614`
    - Inflow from J_2 J2: `1148717038#0`
    - Corridor Exit: `40696223#1` (Electronic City Expressway)
    - Signal State: Active Green (P_2 P2), vehicle count (88), queue ($1\ \text{veh}$1 veh), and speed ($39.0\ \text{km/h}$39.0 km/h).

3.5. Signal Controller Telemetry Panel

- **3-Lamp Traffic Signal Heads (Red, Yellow, Green):**
    - Strictly illuminates only the active light.
    - **Integrity Rule:** If no signal telemetry has arrived, all lamps remain dimmed with an `"Awaiting signal data"` badge, rather than falsely displaying an illuminated red light or default Phase 0.
- **Min/Max Green & Yellow Clearance Safety Shielding:**
    - Enforces minimum green bounds ($15\text{s} - 20\text{s}$15s – 20s) to avoid rapid signal switching and vehicle collisions.
    - Enforces maximum green bounds ($60\text{s} - 75\text{s}$60s – 75s) to avoid starving conflicting side-street traffic.
    - Enforces a mandatory 4s4s yellow clearance phase before switching green movements.
- **Phase Elapsed Display:** Real-time counter of elapsed green time vs. configured safety bounds.

3.6. Emergency Green Wave Preemption

- **Automatic Ambulance Detection:** Sensed as emergency vehicles reach a 250m250m approach threshold.
- **Progressive Priority Lock:** Conflicting approaches are safely transitioned to yellow clearance and red. The arterial green is held consecutively across $J_1 \rightarrow J_2 \rightarrow J_3$ J1 → J2 → J3, forming an unimpeded emergency corridor.
- **Verified Benchmark Traversal:** Reduces ambulance traversal from 41.0s41.0s (baseline) to 33.6s 33.6s (**18.0% faster emergency response**).

3.7. Empirical Controller Benchmark Comparison Panel

- **Single Source of Truth:** Both the comparative bar chart and the detailed metrics table are generated from `/api/experiments/compare`.
- **Honest Provenance & Academic Integrity:**
  - **Fixed-Time Baseline:** Marked as **Implemented (Measured)** with 5-seed empirical SUMO figures.
  - **Coordinated Adaptive Controller:** Marked as **Implemented (Measured)** with 5-seed empirical SUMO figures.
  - **PPO & DQN:** Marked as **Prototype (Eval Pending)**. The neural network architectures (ActorCriticNetwork, Dueling DQN, GAE, replay buffer, action masking) are implemented in `traffic_service/ai/`, but their benchmark metrics are intentionally displayed as blank / em dashes (`—`) until formal multi-seed convergence checkpoints are trained and evaluated on SUMO.
- **Explicit Attribution:** Clearly displays: `Source: Silk Board Corridor (J1 -> J2 -> J3) • 600s Multi-Seed SUMO Benchmark`.

3.8. Dual-Series Emissions & Environmental Analytics

- **Dual Y-Axes Accumulation Curve:**
  - Left Y-Axis: Cumulative CO_2 CO2 emissions in **kg** (teal area).
  - Right Y-Axis: Cumulative Fuel consumed in **Liters** (purple area).
- **HBEFA 3.1 Model:** Driven by the European Handbook on Emission Factors for Road Transport integrated into SUMO.
- **Benchmark Attribution:** Explicitly labeled: `Benchmark (600s run): 20.6% CO₂ reduction vs baseline` (clarifying this is a verified benchmark result, not an assumed live saving).

4. Rigorous Codebase Verification & Test Suite

The system has been verified using automated regression tests:

```
text
============================= test session starts ==============================
platform win32 -- Python 3.13.16, pytest-9.1.1
collected 34 items

tests\test_api.py ............                                           [ 35%]
tests\test_ai_controller.py ............                                [ 70%]
tests\test_communication.py .....                                       [ 85%]
tests\test_metrics.py ..                                                [ 91%]
tests\test_database.py ..                                               [ 97%]
tests\test_websocket.py .                                               [100%]

======================== 34 passed in 0.65s ========================
```

Coverage by Component:

1. : Validates REST endpoints, simulation lifecycle, signal override constraints, decision rationales, and honest controller statuses.
2. : Validates 14-dim observation normalization, PPO/DQN action selection, action masking for min/max green, reward computation, and fallback controller safety.
3. : Validates inter-intersection communication exchange (queue messages, platoon arrival times) between J_1 J1, J_2 J2, and J_3 J3.
4. : Validates waiting times, travel times, throughput, and HBEFA emission calculations.
5. : Validates SQLite persistence of simulation runs, decision logs, and summary metrics.
6. : Validates real-time WebSocket connection handling and JSON message schemas.

5. Official Presentation Q&A and Talking Points

When presenting this project to government officials, traffic police authorities, or university examiners, use these speaking points:

Q1: "Why does the Silk Board junction need coordinated adaptive control rather than isolated smart timers?"

> *"Silk Board (J_2 J2) cannot be solved in isolation. Traffic arriving at Silk Board is primarily released by the Hosur North Feeder (J_1 J1). In our coordinated system, J_1 J1 transmits its platoon departure data to J_2 J2 through inter-intersection communication. J_2 J2 calculates the 400m400m arrival lag and extends green time before vehicles reach the junction, preventing spillback queues and cutting corridor travel time by 36.8%36.8%."*

Q2: "What happens if sensors fail, a camera disconnects, or the AI returns an invalid action?"

> *"Safety is guaranteed by design. The controller features a hard-coded safety shield: minimum green bounds (15s15s), maximum green bounds (75s75s), and a mandatory 4s4s yellow transition clearance are enforced by software constraints. If telemetry is lost or the AI fails, the system automatically defaults to a deterministic queue-pressure fallback controller."*

Q3: "Why are PPO and DQN marked as 'Prototype (Evaluation Pending)'?"

> *"We adhere to research and operational integrity. The neural network architectures (Actor-Critic, Double-DQN, GAE advantage estimator, replay buffer) are implemented in `traffic_service/ai/`. However, we refuse to present preset multiplier estimates as empirical measurements. Until formal multi-seed convergence training checkpoints are evaluated on SUMO, they are transparently labeled as prototypes, while our Coordinated Adaptive Controller has full 55-seed measured verification."*

Q4: "How does the Emergency Green Wave ensure ambulance clearance without causing accidents?"

> *"When an ambulance is detected at the 250m250m approach zone, the preemption module does not instantly cut signals, which could cause rear-end collisions. Instead, it transitions active green phases through a safe 4s4s yellow clearance, turns conflicting movements red, and locks the Hosur corridor green across J_1 J1, J_2 J2, and J_3 J3. This yields an 18.0%18.0% faster traversal time."*